<table>
<tr><th colspan="3" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>9</td><td>(a)</td><td>(i)</td><td>First (pointing) finger points from {N $\rightarrow$ S / left to right / direction of the field} (1)
Thumb points {in direction of motion [of AB] / up} (1)
Second (middle) finger gives [induced] current which is <u>A to B</u> (1)</td><td>1
1</td><td>1</td><td></td><td>3</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td><table>
<tr><th rowspan="2">Change</th><th colspan="3">Effect on voltage</th><th colspan="3">Effect on time for 1 cycle</th></tr>
<tr><th>Decreases</th><th>Stays the same</th><th>Increases</th><th>Decreases</th><th>Stays the same</th><th>Increases</th></tr>
<tr><td>Weaker magnets</td><td>✓</td><td></td><td></td><td></td><td>✓</td><td></td></tr>
<tr><td>Coil spins faster</td><td></td><td></td><td>✓</td><td>✓</td><td></td><td></td></tr>
<tr><td>More turns in the coil</td><td></td><td></td><td>✓</td><td></td><td>✓</td><td></td></tr>
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1 mark for each correct row</td><td></td><td>3</td><td></td><td>3</td><td></td><td></td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td>An <u>alternating</u> magnetic field (1)
is linked to the secondary coil by the iron core (1) OWTTE</td><td>2</td><td></td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Substitution: $\frac{25\,000}{400\,000} = \frac{600}{N_2}$ (1)
Rearrange equation (1)
So secondary turns = 16 $\times$ 600 = 9600 (1)
Award 1 mark for answer of 16 or 0.0625</td><td>1</td><td>1
1</td><td></td><td>3</td><td>3</td><td></td></tr>
<tr><td></td><td></td><td></td><td>Question 9 total</td><td>5</td><td>6</td><td>0</td><td>11</td><td>3</td><td>0</td></tr>
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